

LESSON 2-10 • SESSION 2 • PRACTICE • FORM B

Matching Center and Spread

Learning target – I can match a measure of center and a measure of spread to the shape of a data set. · 6.SP.B.5d

REMEMBER

- Symmetric, no outlier: **mean and MAD**. Skewed or outlier: **median and IQR**.
- **Never mix the pairs**. MAD is built from the mean; IQR is built from the median.
- A long tail to the right means **skewed right**; to the left means **skewed left**.

WORKED EXAMPLE

Set C: 20, 22, 24, 26, 28. Set D: 20, 22, 24, 26, 150. Pick a pair for each.

Step 1 C is symmetric: mean 24, $MAD\ 12 \div 5 = 2.4$.

Step 2 D has an outlier (150): median 24, with the IQR.

Answer C: mean and MAD. D: median and IQR.

WORDS symmetric (*simétrico*) · skewed (*asimétrico*) · mean and MAD (*media y DMA*) · median and IQR (*mediana y rango intercuartil*)

PRACTICE

show your work · label units

1. Write the pair you would report: mean and MAD, or median and IQR.

a. 52, 54, 56, 58, 60 _____

b. 52, 54, 56, 58, 310 _____

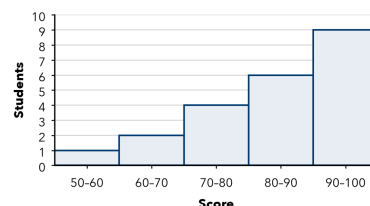
c. 63, 65, 67, 69, 71, 73, 75 _____

d. 1, 45, 47, 49, 51 _____

2. **Test Scores.** The histogram shows the scores of 22 students.

a. Shape: _____

b. Pair to report: _____



3. **Balanced Scores.** Set R is 40, 43, 45, 47, 50. It is symmetric.

Hint Find the mean first. MAD = average distance from it.

a. Mean: _____

b. MAD: _____

Show your work.

4. **One Big Night.** Ticket sales for seven shows were 14, 16, 17, 19, 22, 24 and 350.

Hint Q1 is the middle of the lower half; Q3 of the upper half.

a. Median: _____

b. IQR: _____

Show your work.

5. **Mismatched Pair.** Ana reports a mean of 64 for a symmetric set, with the IQR for spread. What spread should she use?

Hint Which center is the IQR built from?

Show your work.

Spread _____

6. **Book Fair.** Daily sales: 63, 65, 67, 69, 71, 73, 75. Name the pair to report. Find the center.

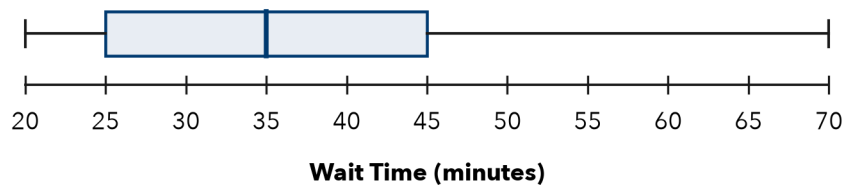
Show your work.

Center _____

7. **One New Value.** The set 60, 62, 64, 66, 68 is symmetric. Then a new value, 400, joins the set.
- a. Pair to report before: _____
 - b. Pair to report after: _____
 - c. New median: _____

Show your work.

8. **Deli Wait Times.** The box plot shows wait times, in minutes.



- a. Shape: _____
- b. Median: _____ min
- c. IQR: _____ min

FIND THE ERROR

explain the mistake, then fix it

9. **Priya’s Work.** Priya saw a set with one huge outlier and chose the mean and MAD, saying “the mean always uses every value.” Explain her mistake.

TEST PRACTICE

choose the best answer

10. A long tail to the right means the data is –
- A symmetric
B skewed right
C skewed left
D clustered
11. An outlier makes you switch from the mean to the –
- A MAD
B range
C total
D median

SELF-CHECK CODE

find your answer · copy its letter into that problem’s box · spell a math word

20 D	19 L	45 B	2.8 A	8 A	69 N	14 Q	65 C	64 Z	35 E
#3a	#3b	#4a	#4b	#6	#7c	#8b	#8c		

CHECK YOURSELF

☐ I can do this on my own
☐ I need a little help
☐ I have a question about #____